

Pipeline for Benchmark Construction

1. Project Scope and Design Principles

This pipeline defines a reproducible methodology for constructing a benchmark composed exclusively of **Core report tasks** and **Set-enumeration tasks**. The Rare-domain track is intentionally excluded from this version of the workflow.

The benchmark is designed around two task families with distinct output regimes:

- **Core tasks** require concise but substantive synthesized reports grounded in multiple sources.
- **Set tasks** require bounded answer sets that can be scored through gold-set matching and precision/recall-style metrics.

The principal design objective is to ensure that every task is traceable to publicly accessible, non-fictional, fact-rich sources published between **2018 and 2025**, and that task construction is sufficiently constrained to minimize ambiguity, hidden leakage, and annotation drift.

Three methodological commitments structure the entire pipeline:

1. **Source-groundedness**: every task must be derivable from the underlying source package without speculative inference.
 2. **Reproducibility**: each source, cluster, task, rubric, and gold set must be versioned and auditable.
 3. **Comparability**: tasks must be sufficiently standardized to allow meaningful model evaluation across domains and difficulty levels.
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2. Phase I – Source Acquisition

2.1 Objective

The first phase is the construction of a high-quality source pool. At this stage, the goal is not yet to create tasks, but rather to assemble candidate documents that satisfy the benchmark’s formal requirements.

2.2 Inclusion criteria

Each candidate source must satisfy all of the following conditions:

- publicly accessible, without requiring a paywall or otherwise available through a legitimate archived copy;
- published between 2018 and 2025;
- contains verifiable factual content, such as numbers, named entities, institutional comparisons, regulatory statements, or documented outcomes;

- sufficiently substantive, either through length (approximately 1,500 words or more) or through equivalent structured content such as tables, registries, annual reports, or official databases;
- not primarily editorial, polemical, or opinion-based.

2.3 Recommended source classes

The source pool should be diversified across the two task families.

For **Core** tasks, suitable source classes include:

- policy and institutional reports;
- annual reports and investor-relations publications;
- long-form analytical journalism;
- survey reports;
- review-style academic or technical syntheses.

For **Set** tasks, suitable source classes include:

- official registries;
- public databases;
- structured lists;
- institutional directories;
- legal or administrative registers;
- Wikipedia list pages when corroborated by more authoritative sources if needed.

2.4 Output of the phase

The output of Phase I is a raw candidate source inventory containing, for each item:

- provisional source identifier;
- title;
- URL;
- publication date;
- source class;
- tentative domain assignment;
- initial notes on factual richness and usability.

3. Phase II – Source Verification and Filtering

3.1 Objective

This phase removes materials that do not satisfy the benchmark’s formal standards. The purpose is to ensure that all downstream tasks are grounded in stable, trustworthy documents.

3.2 Verification steps

Each candidate source is screened for:

- access status;
- publication window compliance;
- factual density;
- domain relevance;
- structural adequacy;
- absence of duplicated or near-duplicated content.

3.3 Relevance to task families

A source is retained only if it can plausibly support at least one task family with minimal distortion:

- **Core** sources must permit synthesis, comparison, or explanation across multiple evidence points.
- **Set** sources must support a bounded answer set whose membership rules can be written explicitly.

3.4 Output of the phase

Each source receives one of four statuses:

- **accepted**;
- **borderline**;
- **rejected**;
- **needs human review**.

Only accepted sources proceed automatically; borderline sources require explicit annotation review.

4. Phase III – Deduplication and Canonicalization

4.1 Objective

The benchmark must avoid accidental redundancy. Duplicate content is problematic not only when URLs are identical, but also when different URLs resolve to the same underlying document or a near-identical version of it.

4.2 Deduplication logic

Deduplication should operate at multiple levels:

1. exact URL matching;
2. canonical URL matching;
3. document identity matching through content hash or near-duplicate detection;

4. manual resolution of ambiguous cases.

4.3 Canonical source representation

Every accepted document family should be mapped to a canonical source record, with supporting aliases or archival variants retained as metadata rather than separate substantive items.

4.4 Output of the phase

The result is a deduplicated source inventory with:

- canonical source IDs;
 - variant URLs;
 - version dates;
 - archive links where applicable;
 - deduplication status;
 - reviewer notes for borderline cases.
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5. Phase IV – Source Normalization

5.1 Objective

Normalization transforms heterogeneous documents into a uniform data structure suitable for annotation, clustering, and task generation.

5.2 Standard source schema

A normalized source record should include at least:

- source ID;
- canonical source ID;
- title;
- URL;
- archive URL;
- publication date;
- language of origin;
- domain;
- source type;
- source subtype;
- access status;
- approximate length or structured-equivalent measure;
- factual density indicator;
- extracted key entities;
- quality notes.

5.3 Family-specific metadata

For **Core** sources, additional metadata should include:

- thematic angle;
- likely comparison dimensions;
- candidate synthesis relevance.

For **Set** sources, additional metadata should include:

- candidate answer-set type;
- entity density;
- boundedness of the expected answer set;
- inclusion/exclusion ambiguity risks.

5.4 Output of the phase

The result is a clean, machine-readable, and human-readable source table that becomes the backbone of the benchmark workflow.

6. Phase V – Extraction and Translation Strategy

6.1 Rationale

Translation should not be treated as a late cosmetic step. It is a structural component of the pipeline because it affects information fidelity, entity stability, and task design.

6.2 Recommended sequence

The recommended order is:

1. select source;
2. extract key passages;
3. translate the extracted passages;
4. store original and translated versions in parallel;
5. use the bilingual packet for annotation and task design;
6. translate the final task prompt only after task logic has been finalized.

6.3 Why this order matters

This order reduces the likelihood of:

- distortion during translation;
- loss of structural cues;
- premature reinterpretation of source meaning;
- prompt drift during task formulation;
- entity-level inconsistency in set tasks.

6.4 Translation principles

The translation layer should prioritize:

- semantic fidelity over stylistic elegance;
- entity-name preservation;
- consistent terminology across sources;
- explicit handling of ambiguous terms;
- lexical stability for repeated entities and concepts.

6.5 Output of the phase

Each source packet should contain:

- original excerpt;
 - translated excerpt;
 - entity glossary;
 - alignment notes;
 - translation confidence notes.
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7. Phase VI – Source Annotation

7.1 Objective

Annotation converts a source from a static document into a benchmark-building asset with explicit task potential.

7.2 Common annotation fields

All sources should be labeled with:

- domain;
- source type;
- difficulty proxy;
- factual richness;
- ambiguity risk;
- task suitability.

7.3 Difficulty proxy

A three-level proxy is sufficient at source stage:

- **Low**: primarily descriptive;
- **Medium**: analytically structured;
- **High**: multi-factor or potentially ambiguous.

7.4 Family-specific annotation

For **Core** sources, annotators should note:

- comparison dimensions;
- synthesis potential;
- likely report themes.

For **Set** sources, annotators should note:

- entity boundaries;
- set exhaustiveness potential;
- likely distractor patterns;
- canonical naming issues.

7.5 Output of the phase

The output is an annotated source register that supports clustering and task design.

8. Phase VII – Source Clustering into Task Units

8.1 Objective

Clustering is the intermediate step that links the source pool to concrete tasks. A cluster is a deliberately chosen package of one or more sources sharing a common future task logic.

8.2 Cluster composition

For **Core** tasks, clusters typically include **two to three sources**. This supports multi-source synthesis and prevents overreliance on a single document.

For **Set** tasks, clusters typically include **one primary source** and, where useful, **one supporting source**.

8.3 Cluster design criteria

A strong cluster should exhibit:

- coherent topical focus;
- compatible evidentiary structure;
- a clear intended output regime;
- manageable ambiguity;
- minimal internal contradiction.

8.4 Cluster metadata

Each cluster should store:

- cluster ID;

- family assignment;
- subtype;
- source IDs;
- cluster rationale;
- expected output type;
- anticipated difficulty.

8.5 Output of the phase

The result is a set of task-ready source clusters that can be converted into prompts and evaluation artifacts.

9. Phase VIII – Task Family Assignment

9.1 Objective

Each cluster must be assigned to one of the benchmark’s task families based on the expected output structure and reasoning mode.

9.2 Assignment rule

- If the output is a synthesized narrative or comparison, the cluster becomes a **Core** task.
- If the output is a bounded list or set, the cluster becomes a **Set** task.

9.3 Subtype assignment for Set

Set tasks may additionally be classified into subtypes such as:

- entity-list;
- filtered multi-hop;
- comparison-selection.

9.4 Output of the phase

Each cluster receives a finalized family label and subtype label, which determine the downstream rubric or gold-set design.

10. Phase IX – Prompt Construction

10.1 Objective

Prompt construction turns the cluster into an executable benchmark item.

10.2 Structural template

Each prompt should contain:

- **Situation:** the motivating context;
- **Scope:** domain, geography, time frame, and language;
- **Task:** the explicit instruction;
- **Output format:** report, list, or table;
- **Constraints:** evidence requirements, comparison requirements, and prohibition of unsupported claims.

10.3 Core prompt design

Core prompts should:

- require synthesis rather than extraction;
- encourage comparison where relevant;
- define the target audience implicitly or explicitly;
- avoid over-constraining the response to a single citation pattern;
- keep the task answerable from the provided sources without external assumptions.

10.4 Set prompt design

Set prompts should:

- state the membership condition explicitly;
- define the answer space as bounded;
- avoid linguistic ambiguity in inclusion criteria;
- require a list, table, or structured set;
- make boundary conditions visible to the evaluator.

10.5 Output of the phase

The output is a draft task prompt that is aligned with the family-specific evaluation logic.

11. Phase X – Evaluation Artifact Construction

11.1 Core rubrics

Core tasks require rubrics with five fixed criteria:

1. Coverage;
2. Accuracy;
3. Reasoning;
4. Use of evidence;
5. Clarity and structure.

Each criterion is scored on a 0–2 scale.

11.2 Core rubric development procedure

For each Core task:

1. extract 5–10 key facts from the source bundle;
2. identify the expected response sections;
3. specify positive and negative examples for each criterion;
4. record common failure modes;
5. add notes on borderline cases.

11.3 Set gold sets

Set tasks require a gold set rather than a narrative rubric. Each gold set should include:

- correct items;
- acceptable variants;
- optional distractors;
- explicit exclusion rules;
- boundary notes.

11.4 Set evaluation logic

Set tasks are evaluated through:

- precision;
- recall;
- F1.

The gold set must therefore be exhaustive or clearly bounded enough to support stable scoring.

11.5 Output of the phase

The result is a task-level evaluation artifact: either a rubric ID for Core or a gold-set ID for Set.

12. Phase XI – Human Review and Adjudication

12.1 Objective

Human review is required to control quality before the benchmark is frozen. The review process should be staged rather than monolithic.

12.2 Review stages

A robust review process includes:

1. **creator self-check**;
2. **second-annotator review**;
3. **adjudication** for disagreements.

12.3 Review criteria

Reviewers should verify:

- source adequacy;
- task answerability;
- prompt clarity;
- family assignment correctness;
- rubric or gold-set completeness;
- difficulty label plausibility;
- translation fidelity;
- absence of leakage or hidden ambiguity.

12.4 Output of the phase

The result is a validated task item with resolved disputes and documented reviewer notes.

13. Phase XII – Pilot Testing with Models

13.1 Objective

Pilot testing checks whether the benchmark meaningfully differentiates model behavior and whether task design is robust in practice.

13.2 Pilot protocol

The pilot should include a small but diverse subset of tasks from both Core and Set. A reasonable pilot uses multiple model classes, for example:

- a weaker baseline model;
- a stronger general-purpose model;
- a model with retrieval support, if available;
- optionally, a second strong model for comparison.

13.3 What to measure

For Core tasks:

- coverage of major points;
- factual correctness;
- quality of comparisons;
- evidence use;
- structural coherence.

For Set tasks:

- precision;

- recall;
- F1;
- over-generation;
- under-generation;
- boundary confusion.

13.4 Diagnostic questions

Pilot testing should determine whether:

- tasks are too easy;
- tasks are too hard;
- prompts leak the answer;
- translation changed task semantics;
- the rubric or gold set needs refinement;
- model rankings are stable enough to be meaningful.

13.5 Output of the phase

The output is an error-analysis report with task-level and model-level findings.

14. Phase XIII – Revision and Stabilization

14.1 Objective

No benchmark should be frozen before it has been revised in response to pilot findings.

14.2 Typical revision targets

Revision may be required for:

- ambiguous scope statements;
- weak source bundles;
- overly narrow or overly broad prompts;
- incomplete gold sets;
- under-specified rubrics;
- translation-induced meaning shifts;
- duplicate or overlapping tasks.

14.3 Revision principle

The goal is not to make the benchmark easier or harder in an ad hoc way, but to make the benchmark more valid, more discriminative, and more internally consistent.

14.4 Output of the phase

A stabilized task inventory ready for freeze.

15. Phase XIV – Dataset Freeze and Version Control

15.1 Objective

The benchmark must be frozen once validation is complete.

15.2 Freeze package

The frozen release should include:

- source inventory;
- source annotations;
- cluster assignments;
- final prompts;
- rubrics or gold sets;
- translation memory;
- review logs;
- pilot-test results;
- change log and version tags.

15.3 Post-freeze policy

After freeze, substantive content should not change without a new version identifier. This prevents silent benchmark drift.

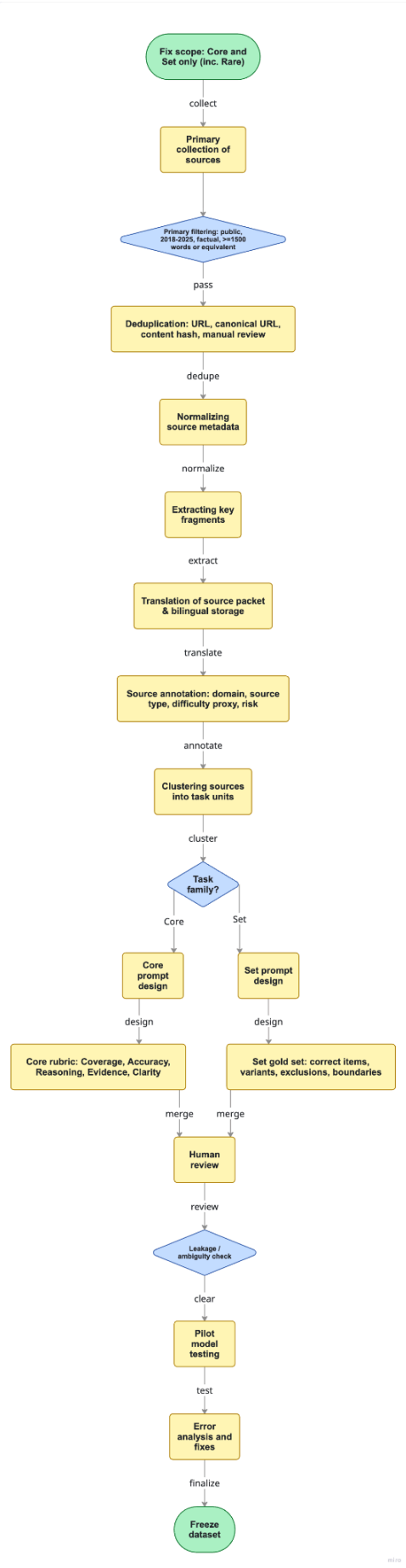
16. Recommended Operational Sequence

A concise operational order is as follows:

1. define scope;
2. collect candidate sources;
3. verify eligibility;
4. deduplicate;
5. normalize metadata;
6. extract and translate source excerpts;
7. annotate sources;
8. cluster sources into task units;
9. assign Core or Set family;
10. draft prompts;
11. construct rubrics or gold sets;
12. conduct human re
13. view;
14. run pilot model evaluations;

15. revise as needed;
16. freeze the dataset.

17. High-Level Flowchart



18. Summary

This pipeline operationalizes benchmark development as a staged and auditable process rather than as a single prompt-writing exercise. The central methodological distinction is between the two task families: Core tasks require multi-source synthesis and rubric-based evaluation, whereas Set tasks require bounded answer sets and gold-set evaluation. Translation is treated as a structural stage rather than a cosmetic one, and model testing is positioned before final freeze so that construction flaws can be corrected while the benchmark is still mutable.